



Brecksville-
Broadview Heights
City School District

Brecksville-Broadview Heights City School District

Curriculum Framework

Brecksville-Broadview Heights City Schools
6638 Mill Rd
Brecksville, Ohio 44141

Honors Design Team 2025

District Leadership

Superintendent of Schools :: Jeff Harrison
Assistant Superintendent of Schools :: Brian Wycuff
Director of Teaching and Learning :: Dr. David Martin

Board of Education

Mark Dosen :: President
Ellen Kramer :: Vice President
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Tish Kwiatkowski
Eva O'Mara

Acknowledgment

This document represents a strong effort on the part of the following individuals:

- Ryan Goubeaux



**Brecksville
Broadview Heights
City School District**

who we are. where we are going. what we value.

OUR MISSION

The BBHCSD exists to rigorously educate our students to the highest standards, to boldly empower our school family with exciting learning experiences, and to collaboratively engage our community to cultivate productive future citizens of a global society.

OUR VISION

The BBHCSD will be the destination district by pushing the boundaries of academic excellence and expanding innovative opportunities for students, while advocating for all, in a safe and accountable educational environment.

OUR VALUES

- We value academic excellence and providing innovative learning opportunities for all
- We value the finest educators and staff to empower our students
- We value school safety and security in protocol, thought, and action to create a welcoming school climate
- We value transparent, engaging, and collaborative communications
- We value responsible stewardship of the dynamic resources in the BBHCSD
- We value the comprehensive preparedness of our students' future success

OUR CULTURE

The BBHCSD will encourage all to Bee Your Best, Bee The Future, and Bee One Community, and will accomplish this by building safety, sharing vulnerability, and establishing a shared purpose with the values above.

Together, we will ***Protect the Banner of the BBHCSD.***

Instructional & Curricular Philosophy

The curriculum and instruction of the Brecksville Broadview Heights City School District will provide an educational program to assist all students to realize their potential to make a positive impact as contributing members of a global society. Both the curriculum and the instructional process reflect that each student is a unique individual and learner.

Students are prepared for their futures through strong curriculum and instructional practices instilling the competencies of Emotionally-Prepared, Future-Ready, Globally-Responsible, and Self-Reliant. In addition, all learners understand the outcomes of Always Improving, Stronger Together, and Leave a Legacy stated in the district's culture playbook. Additionally, the curriculum and instruction aims to achieve with all learners the Culture Playbook outcomes Always Improving, Stronger Together and Leave a Legacy.

Students are at the center of all we do in the Brecksville Broadview Heights City School District. We practice standards-driven curriculum and instruction in all content areas:

- We anchor all learning in Ohio's learning standards.
- We utilize data-driven, evidence-based instruction and decision making.
- We utilize assessments directly aligned to Ohio's learning standards to collect data on student growth and mastery, adjusting and adapting instruction to meet the learning needs of all students.
- We foster a culture of continuous growth in learning for both students and staff.
- We believe that learning is a lifelong pursuit and that everyone has the capacity to grow and continually learn.
- We provide supportive learning environments that encourage active engagement, curiosity, exploration, and rigorous levels of learning.



EDUCATE.
EMPOWER.
ENGAGE.

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ATTRIBUTES OF A BEE

FUTURE READY

BBH students will become lifelong learners and acquire career knowledge along with core technical skills. Bees will be independent thinkers, problem-solvers and possess communications and listening skills.

EMOTIONALLY PREPARED

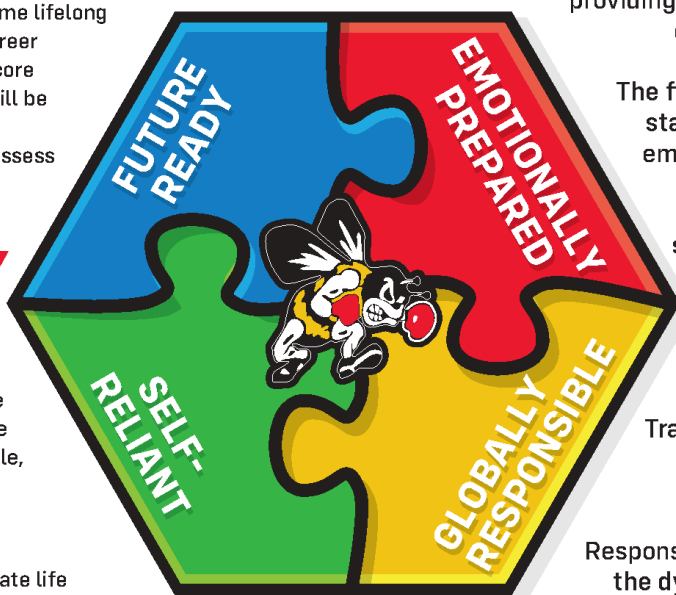
Bees will be self-aware and self-confident and have a growth mindset, while being resilient, adaptable, and flexible.

SELF-RELIANT

Bees will have appropriate life skills, be resourceful, financially savvy, and be self-directed.

GLOBALLY RESPONSIBLE

Our graduates will be engaged citizens, team collaborators, strong stewards of the environment, with broad horizons.



OUR COMMUNITY'S SCHOOL VALUES

Academic excellence and providing innovative learning opportunities for all

The finest educators and staff to constructively empower our students

School safety and security in protocol, action, thought in creating a welcoming school climate

Transparent, engaging and collaborative communications

Responsible stewardship of the dynamic resources in the BBHCSD

Comprehensive preparedness of our students' future college and career pathways

Protecting the Banner of the BBHCSD!

Honors Design Team

District Overview of Content

The International Society for Technology in Education (ISTE) standards serve as a cornerstone to integrating technology within educational settings, aiming to prepare students adeptly for the digital age's complexities. Central to these standards is the empowerment of learners, encouraging students to proactively engage with their education through technological tools to set and achieve personal learning goals. This is complemented by fostering a sense of Digital Citizenship, where students learn to navigate the digital world responsibly, understanding their rights, responsibilities, and the importance of ethical online behavior. Such foundational skills are pivotal in nurturing informed individuals capable of thriving within a globally connected society.

ISTE standards champion the roles of students as Knowledge Constructors and Innovative Designers, advocating for an active interaction with digital resources to deepen knowledge and creatively address problems. This pedagogical approach not only promotes the acquisition and application of knowledge through technology but also stimulates innovative thinking and problem-solving skills critical in a tech-centric world. Additionally, the standards emphasize Computational Thinking and Creative Communication, equipping students with the skills to systematically solve problems and express ideas effectively across digital mediums. Together with fostering global collaboration, these standards aim to develop individuals who are not just technologically proficient but can also creatively, ethically, and collaboratively apply their skills across global platforms, embodying a comprehensive approach to STEM education.

ISTE Standards for Students overview is paraphrased from ISTE

Course Overview

Honors Design Team is an advanced high school course that provides students with a college-level experience in design and project management. This course is designed for students who have a keen interest in a multitude of disciplines and are eager to explore real-world problem-solving through hands-on projects. Throughout the course, students will work both individually and in teams to brainstorm, design, build, and test solutions to authentic real world challenges.

Course Objectives:

1. To develop students' skills in design, problem-solving, and critical thinking.
2. To foster collaboration, communication, and teamwork among students.

3. To provide practical experience in project management, including planning, budgeting, and time management.
4. To encourage creativity and innovation in addressing real-world problems.
5. To introduce students to tools and technologies commonly used in the field.
6. To cultivate a mindset of continuous improvement and iteration in the design process.

Course Content:

1. Advanced use of Design: Understanding the design process, problem identification, and problem statement formulation.
2. Research and Brainstorming: Conducting research, gathering information, generating ideas, and evaluating potential solutions.
3. Design Development: Conceptualizing designs, creating sketches, drafting plans, and utilizing computer-aided design (CAD) or graphic software.
4. Prototyping and Fabrication: Building physical prototypes using various materials, tools, and manufacturing techniques.
5. Testing and Evaluation: Conducting experiments, collecting data, analyzing results, and refining designs based on testing outcomes.
6. Project Management: Planning project timelines, allocating resources, managing budgets, and coordinating team activities.
7. Presentation and Documentation: Communicating design ideas effectively through presentations, technical reports, and documentation.
8. Ethical and Social Considerations: Examining the ethical implications of engineering solutions and considering societal impacts.
9. Final Project Showcase: Presenting completed projects to peers, teachers, and invited guests. Reflecting on the design process and lessons learned.

Materials

1. Textbooks and reference materials on engineering design principles and methodologies.
2. Computer-aided design (CAD) software such as AutoCAD, Fusion 360, Revit, Lightburn...
3. Computer-aided graphic design software such as Illustrator, Photoshop, Canva...
4. Basic hand tools, measuring instruments, and prototyping materials (e.g., wood, plastic, metal).
5. Access to a workshop or makerspace equipped with fabrication tools (e.g., 3D printers, laser cutter, CNC machines).

Assessments

Assessment in Honors Design Team will be based on a combination of individual and team-based assignments, projects, presentations, tests and portfolio. Evaluation criteria may include creativity, technical proficiency, teamwork, communication skills, project management, and adherence to deadlines. Additionally, students will be encouraged to self-assess their progress and participate in peer evaluations.

Content Standards

[ISTE Standards for Students](#)

Scope and Sequence of Knowledge and Skills to Be Learned

Quarter 1-4

Review and further analysis of Technology Literacy

Review and further analysis of Design Process method of problem solving

Review and further analysis of Machine Safety and operation procedures

Follow Design Process to create solution to desired problem (repeat for each project)